



**Verified Carbon
Standard**

15MW FLOATING SOLAR PV PROJECT IN HIMACHAL PRADESH



Document Prepared by Kosher Climate India Private Limited

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Prepared by	Kosher Climate India Private Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The purpose of this project is to generate electricity by harnessing the solar energy by using of solar photovoltaic technology and there by feed the generated electricity to the Indian national grid. The Solar photovoltaic based Panels converts the available solar radiation into the DC power and there by installed the Inverters to convert the DC power to the AC Power.

SJVN Green Energy Limited (A Wholly Owned Subsidiary of SJVN limited) to develop a 15 MW floating solar PV plant on the surface of Nangal Pond, located near Neilla Village, Bilaspur district in the Indian state of Himachal Pradesh.

The project was awarded to SJVN Limited by Solar Energy Corporation of India Ltd. (SECI) through competitive bidding process of SECI. Power generated from the projects is evacuated through national grid of India to be sold to Bhakra Beas Management Board (BBMB). The Power Purchase Agreement (PPA) has been executed between SJVN Green Energy Limited and Bhakra Beas Management Board (BBMB).

The scenario existing prior to the implementation of the project, is electricity delivered to the grid by the project that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

Project activity is expected to generate annual average electricity of 30,583 MWh and reduces an annual average emission reductions equivalent to 28,601 tCO₂e and 200,204 tCO₂e over the crediting period of 7 years.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	14-September-2024 to 13-September-2031	VCS	Enviance Services Private Limited	07

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Scope: 01 - Energy (renewable / non-renewable)
Project activity type	Type I – Renewable Energy Projects

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.3 of VCS Standard version 4.6, table no.1 excluded project activities (The exclusion does not apply to concentrated solar thermal to electricity, floating solar PV, or energy storage systems e.g., batteries).

The project activity applies the small-scale CDM approved methodology AMS-I.D: Grid connected renewable electricity generation, Version 18.0, which is eligible under VCS GHG program. Further, this project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.

Thus, the proposed project is eligible under the scope of the VCS program.

1.4.2 AFOLU project eligibility

This is a non-AFOLU project. This section is not applicable.

1.4.3 Transfer project eligibility

This project is not transferred from any other GHG programs and it is not a CPA project.

Hence, this section is not applicable.

1.5 Project Design

Indicate if the project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

The Project is not a Grouped project. This section is Not Applicable.

1.6 Project Proponent

Organization name	SJVN Green Energy Limited
Contact person	Pushkar Verma
Title	DGM/EIC

Address	SJVN Corporate Head Quarters, Shakti Sadan, Shimla - 171006
Telephone	+91 94184 50295
Email	pushkar.verma@sjvn.nic.in

1.7 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna Manchikalapudi
Title	Founder & Managing Director
Address	Zee Plaza, No.1678, Ground Floor, 27th Main Road, Sector 2, HSR Layout, Bengaluru, Karnataka 560102
Telephone	+91 99453 43475
Email	vamsi@kosherclimate.com narendra@kosherclimate.com

1.8 Ownership

SJVN Green Energy Limited is the legal owner of the project activity. Respective project commissioning certificate and the Power Purchase Agreement of the project would be the supporting document to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.6) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. Also, other legal compliances may be considered.

1.9 Project Start Date

Project start date	14-September-2024 (Expected)
Justification	As per the section 3.8 of the VCS standard v4.6, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. Since the project is not yet started the commercial operations, the expected start of commercial operations has been considered as the project start date.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	14/09/2024 to 13/09/2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

The table below shows the Estimated GHG emission reductions or removals (tCO₂e) for the fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
14-Sep-2024 to 31-Dec-2024	8,700
01-Jan-2025 to 31-Dec-2025	28,989
01-Jan-2026 to 31-Dec-2026	28,844
01-Jan-2027 to 31-Dec-2027	28,700
01-Jan-2028 to 31-Dec-2028	28,556
01-Jan-2029 to 31-Dec-2029	28,414
01-Jan-2030 to 31-Dec-2030	28,272
01-Jan-2031 to 13-Sep-2031	19,729
Total estimated ERRs during the first or fixed crediting period	200,204
Total number of years	7
Average annual ERRs	28,601

1.12 Description of the Project Activity

The purpose of this project is to generate electricity by harnessing the solar energy by using of solar photovoltaic technology and there by feed the generated electricity to the Indian national

grid. The Solar photovoltaic based Panels converts the available solar radiation into the DC power and there by installed the Inverters to convert the DC power to the AC Power.

SJVN Green Energy Limited (A Wholly Owned Subsidiary of SJVN limited) to develop a 15 MW floating solar PV plant on the surface of Nangal Pond, located near Neilla Village, Bilaspur district in the Indian state of Himachal Pradesh.

The scenario existing prior to the implementation of the project, is electricity delivered to the grid by the project that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

The generated electricity from the project is supplied to the Indian National grid. Project Owner has signed a Power Purchase Agreement with MPPMCL to supply the generated solar power at a fixed price for a period of 25 years.

Capacity (MW)	Purpose	Commissioning date (COD)	Village
15	Sale to BBMB	14/09/2024 (Expected)	Neilla

Project activity is expected to generate annual average electricity of 30,583 MWh and reduces an annual average emission reductions equivalent to 28,601 tCO₂e and 200,204 tCO₂e over the crediting period of 7 years.

Technologies/measures:

Project used Crystalline Photovoltaic technology which converts the solar radiation into the electricity. The solar PV plant has the PV modules, Central Inverters, Transformers and other relay and protection systems.

Technical specifications of the major components of the project activity are given below. Solar PV modules were Seasonal tilt mounted and overall life of the projects are 25 years.

Parameters	15 MW Floating Solar
Project Capacity	Capacity- 15 MW (AC)
Technology	Bifacial crystalline Module
Module Make	PREMIER ENERGY
Capacity	545Wp
Total Number of Modules	40,417
Inverter Make	SUNGROW
Inverter	Capacity - 4.4 MW & 3.125 MW
	Model - SG4400UD-20 & G3125HV-32
	Number of Inverters - 4 Nos.
Inverter Transformer Make	DANISH
Capacity	8.75 MVA & 6.25 MVA
Mounting Structure	High Density Polyethylene (HDPE) Floats
Lifetime of the project	25 Years

Solar PV modules convert the available solar radiation into the DC power. Installed Central Inverters will convert the generated DC power into the 660 V AC Power. Total AC power from all the inverter blocks will be pooled into the common switchyard. Total Power will be stepped up to 33 kV in the switchyard by using step-up transformers and transmitted to the nearest BBMB substation.

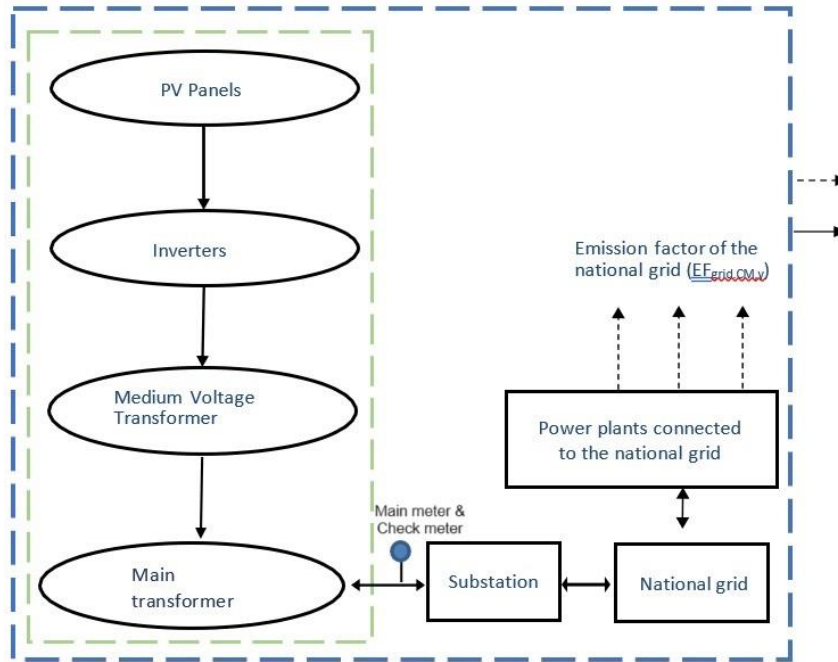


Figure 2: Solar power plant flow diagram

An intelligent automatic monitoring and alarm system (SCADA-Supervisory Control and Data Acquisition) has been already installed in the project control room which will monitor and record the real time data from the plant and alerts the staff in case of any malfunctioning in the equipment operation. However, Separate Energy meters will be installed by the electricity utility to record the import and export of electricity from the plant.

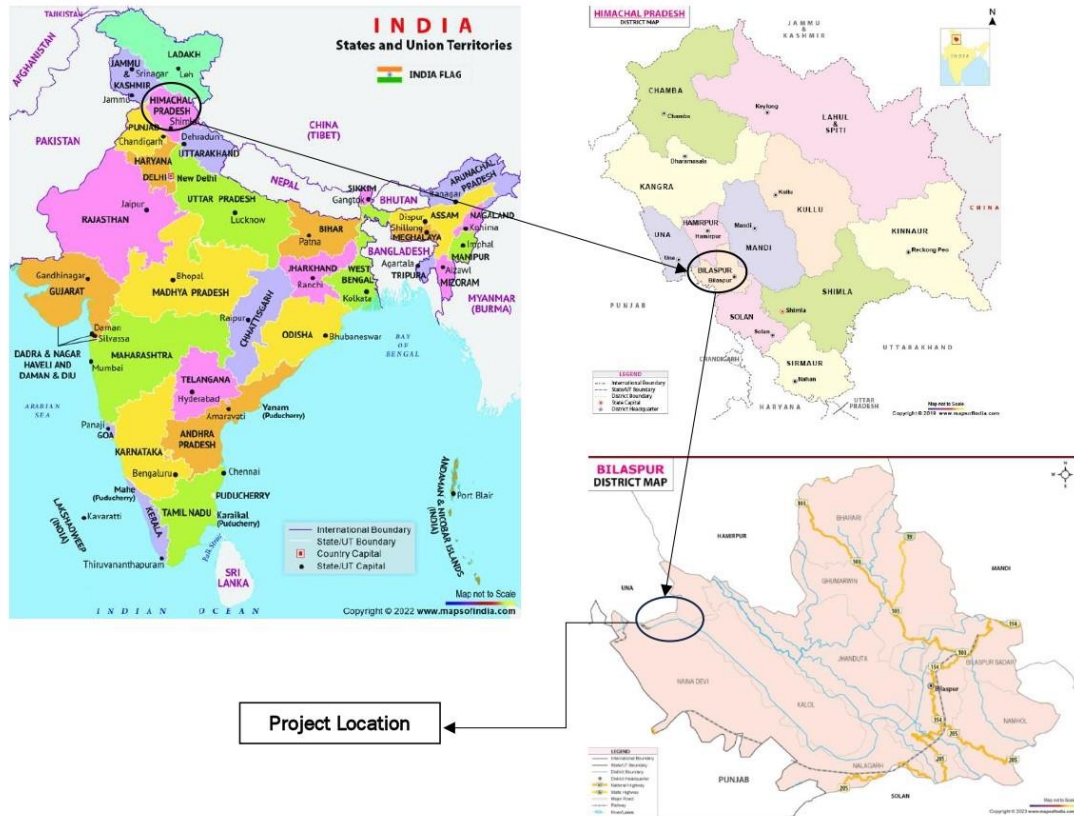
This is a Greenfield project activity generating the electricity from the solar energy and supplying it to the national grid. In the baseline scenario the equivalent of electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07: "Tool to calculate the emission factor for an electricity system", version 7.0. There is no technology transfer occurred in the project activity.

1.13 Project Location

Address and geographic coordinates of the physical site of the project activity			
Project	Physical address	Latitude	Longitude

15 MW Floating Solar Project	Nangal Pond, Near Village Neilla, Bilaspur, Himachal Pradesh, India	31°24'50.97"N (31.4141° N)	76°23'48.35"E (76.3968° E)
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The Project is located in the state of Himachal Pradesh, India.



1.14 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the project activity, is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool 07: Tool to calculate the emission factor for an electricity system”, version 7.0. This is a Greenfield project activity. There was no activity at the site of the project owner prior to the implementation of this project activity. Hence pre-project scenario and baseline scenario is the same.

Refer Section 3.4 (Baseline Scenario) for detailed understanding of the same.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project activity conforms to all the applicable laws and regulations in India:

- Power generation using renewable energy is not a legal requirement or a mandatory option.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation

There is no mandatory requirement to implement the project activity. The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the “Consent to Operate” for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations.

As per the paragraph no 3.14.1 of the VCS standard v4.4 the project conforms to all the applicable laws and regulations and is not implemented by the force of law. This is a voluntary activity undertaken by the project owner in compliance with all the legal requirement in the host country.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?

☐ Yes

☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes

☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes

☐ No

1.18 Sustainable Development Contributions

The project is expected to contribute 03 SDGs which are SDG 7, 8 and 13.

SDG 7 Energy: The project contributes SDG Target 7.2 “By 2030, increase substantially the share of renewable energy in the global energy mix” by the utilization of solar power as a renewable energy source.

SDG 8 Economic Growth: The project creates short- and long-term employment opportunities for both men and women during construction and operation phases of the project, so it contributes to SDG Target 8.5 “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value”.

SDG 13 Climate Change: The project produces clean renewable energy by diminishing CO2 emissions. Therefore, it contributes SDG Target 13.2 “Integrate climate change measures into national policies, strategies and planning.”

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

There is no leakage involved in the project. Thus, Leakage management is not required in this case.

1.19.2 Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not Applicable

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders for the project have been prioritized by identifying direct (those who have a direct interest or influence on the project) and indirect stakeholders whose interest is indirect. Stakeholders has been identified based on potential impact from all related facilities associated with the project. Hence, the stakeholders identified are Local people, employees, Local authorities, drivers, etc.
Legal or customary tenure/access rights	The legal and customary rights of the water body during and post implementation of the project activity lies with Bhakra Beas Management Board (BBMB). Project proponent have no rights on the legal ownership of the water body.

	There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Stakeholder diversity and changes over time	Each community has its own cultural practices, customs, and beliefs, contributing to the overall cultural heritage and integrity of the region. The project anticipates no significant changes to the group's cultural integrity in the project region. Further, the project activities by contributing to the environmental, economic, and social benefits, play a crucial role in enabling local communities to remain located in the project area.
Expected changes in well-being	Environmental well-being: Solar power is one of the cleanest renewable energy powers and does not involve any fossil fuel. There are no GHG emissions. The impact on land, water, air and soil is negligible. Thus, the project activity contributes to environmental well-being without causing any negative impact on the surrounding environment. Economic well-being: The project activity generates permanent and temporary employment opportunity within the vicinity of the project. The electricity supply in the nearby area improves which directly and indirectly improves the economy and life style of the area.
Location of stakeholders	All community members, those who are directly or indirectly impacted by the project are mostly local people who reside near the project areas.
Location of resources	The location of territories and resources to which stakeholders have customary access is the nearby land and Nangal Pond in Bilaspur District of Himachal Pradesh.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	17-February -2024
Stakeholder engagement process	The LSC invitations were put on public notice to the local people. The consultation attendees were given sufficient time and scope to discuss the project and clarify their doubts related to the project

	and provide their feedbacks on the project.
Consultation outcome	The PP has conducted a discussion meeting on the project particulars with stakeholders. These sessions ensured unbiased dissemination of all project-related information to stakeholders, irrespective of caste, religion, gender, race, etc., This approach aimed to eliminate any potential barriers hindering the participation of diverse stakeholder groups. The discussion covered a wide range of topics, including project design, location, objectives, costs, benefits, risks, duration, scope, size, impacts, and the carbon project cycle, encompassing validation and verification processes. Based on this information, interested stakeholders provided their inputs through open discussions during the meeting. No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were positive about the project, no further action is required. There were no further comments raised by the stakeholders and they were totally in support for setting up of these kinds of projects in the region.
Ongoing communication	A dedicated email address as well as the phone number and the address of PP were provided to all stakeholders to foster open and accessible communication. A grievance register will be maintained at the site to register the complaints of the stakeholders. There will be an active presence of the Project Proponent team in the project area. They will work closely with the community and other stakeholders to address any queries or grievances in relation to the project. Also assured that a grievance register is always made available at the project site to register their complaints if any in the register and same will be addressed and resolved in the earliest possible time. All communication and consultation shall be performed in a culturally appropriate manner, including language and gender sensitivity.
Stakeholder input	No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The legal and customary rights of the water body during and post implementation of the project activity lies with Bhakra Beas Management Board (BBMB). Project proponent have no rights on the legal ownership of the pond. There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Outcome of FPIC	The legal and customary rights of the water body during and post implementation of the project activity lies with Bhakra Beas Management Board (BBMB). The project proponent has not done encroachment and relocated people, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	The project team devised a structured process in a culturally appropriate manner for receiving input and addressing grievances, by maintaining a grievance register at the site to register the complaints of the stakeholders.
Grievance redress procedure	There is a mechanism in place to address the grievances and resolve any potential conflict between the project proponent and local stakeholders. The project proponent contact information like e-mail and phone numbers are provided to the stakeholders in case they want to contact and share their ideas and convey their grievances.

2.1.5 Public Comments

The Section will be updated post the public comment period.

2.2 Risks to Stakeholders and the Environment

Risks identified	Mitigation or preventative measure taken
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Risks to stakeholder participation	No risks identified	The project team has explained about the project to the stakeholders including project benefits, costs, risks, rights, laws and opportunities on employment generation. The stakeholders are voluntarily willing to participate in the project because of benefits like safeguarding the communities from environmental challenges and various other SDG benefits of the project.
Working conditions	No risks identified	The project has taken comprehensive measures to ensure providing safe and healthy working conditions for its employees.
Safety of women and girls	No risks identified	The women are provided employment opportunities in the project activity with attention given to addressing gender-specific safety concerns. The organization does not tolerate harassment for any women employees which includes sexual, psychological or any other form of harassment that can be a form of discrimination.
Safety of minority and marginalized groups, including children	No risks identified	The project proponent has protocols aimed at safeguarding women, children and other marginal or vulnerable groups.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risks identified	The project activities do not involve generation of pollutants.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	The organization has policies for Non-Discrimination and for the prevention of sexual harassment and would take appropriate steps for preventing such activities at workplace and put in place an appropriate mechanism of complaint/grievance redressal.
Management experience	The Project Proponent has the operational expertise, management strengths, financial capabilities and commitment to quality. They have expertise and experience in implementing similar project activities and engaging communities.
Gender equity in labor and work	The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in Women's Rights Protection Law to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. During stakeholders' consultation process, comments were collected from the local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in Labor Law and Women's Rights Protection Law.
Human trafficking, forced labor, and child labor	The Project will not use victims of human trafficking, forced labor, and child labor. Human trafficking is a serious crime that is strictly regulated and combated in the Indian legal system. India has promulgated the "Labor Law" to protect the legitimate rights and interests of workers and prohibit forced labor. India has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the solar power plant. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.

2.3.2 Human Rights

The PP emphasis on aligning the organizational practices with the human rights outlined in national and regional regulations. All customary rights of the water body and its resources are held by the Bhakra Beas Management Board (BBMB). The project respects protection of the rights of local communities and customary rights holders, aligning with applicable international human rights law, the United Nations Declaration.

2.3.3 Indigenous Peoples and Cultural Heritage

The project does not impact and protects cultural heritage as part of project activities.

2.3.4 Property Rights

Rights to territories and resources	The legal and customary rights of the water body during and post implementation of the project activity lies with Bhakra Beas Management Board (BBMB). Project proponent have no rights on the legal ownership of the pond. There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Respect for property rights	The project respects all stakeholder property rights. The Project Proponent will not alter or impact any of their rights, thus preserving and protecting them.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.
Summary of the benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.
Approval and dissemination of benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	The project does not have any impact on biodiversity and ecosystems.

Soil degradation and soil erosion	No risk identified	Usage of land is very minimal in the project, as it is a floating solar project.
Water consumption and stress	No risk identified	The project involves water conservation measures to ensure the plant consume very minimal water.
Usage of fertilizers	No risk identified	There is no usage of fertilizers involved.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

N/A.

2.4.3 Ecosystem conversion

As the project is not a ARR, ALM, WRC or ACoGS project, this section is not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-I.D	Grid connected renewable electricity generation	18.0 ²
Tool	07	Tool to calculate the emission factor for an electricity system	7.0 ³
Tool	20	Assessment of debundling for small-scale project activities	4.1 ⁴

² <https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI90DBC>

³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-20-v4.1.pdf>

Tool	21	Demonstration of additionality of small-scale project activities	13.1 ⁵
Tool	27	Investment analysis	13.0 ⁶

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-I.D	This methodology is applicable to grid-connected renewable power generation project activities that: (a) install Greenfield power plant; (b) involve a capacity addition to (an) existing plant(s); (c) involve a retrofit of (an) existing plant(s)/unit(s); (d) involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) involve a replacement of (an) existing plant(s)/unit(s)	The project is a Greenfield grid-connected solar power plant. Therefore, it confirms to the said criteria (a).
	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	The project is the installation of a Greenfield solar power plants/units. Therefore, the said criterion is not applicable.
	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a	The project is the installation of a Greenfield grid connected 15 MW

⁵ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-21-v13.1.pdf>

⁶ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v13.pdf>

	small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW	renewable solar power project. Hence this criterion is not applicable.
	Combined heat and power (co-generation) systems are not eligible under this category.	The project is the installation of a Greenfield grid connected 15 MW renewable solar power project. Hence this criterion is not applicable.
	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct ⁷ from the existing units.	The project is the installation of a Greenfield grid connected 15 MW renewable solar power project. Hence this criterion is not applicable.
	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	The project is the installation of a Greenfield grid connected 15 MW renewable solar power project. Hence this criterion is not applicable.
	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal	The project is the installation of a Greenfield grid connected 15 MW renewable solar power project. Hence this criterion is not applicable.

⁷ Physically distinct units are those that are capable of generating electricity without the operation of existing units, and that do not directly affect the mechanical, thermal, or electrical characteristics of the existing facility. For example, the addition of a steam turbine to an existing combustion turbine to create a combined cycle unit would not be considered “physically distinct”.

	energy production with or without electricity” shall be explored.	
	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply	The project is the installation of a Greenfield grid connected 15 MW renewable solar power project. Hence this criterion is not applicable.
Tool07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The Project is a Greenfield solar power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system” version 7.0.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option II a is chosen the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power	Since the project is Greenfield grid -connected solar power project, this condition is applicable and the emission factor has been calculated accordingly. Combined margin grid emission factor has been calculated as per the CO₂ emission factor data base published by the CEA in which for the calculation of

	plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	emission factor, CEA have only considered grid connected plants.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The Project is located in India, a non-Annex I country. Therefore, this tool is applicable for the project activity
	Under this tool, the value applied to the CO ₂ emission factor of bio fuels is zero	The project is a grid connected solar power project and therefore, this criterion is not applicable for the project activity.
Tool21	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board	The project owner is not proposing any new methodology and applied this tool for the demonstration of additionality in compliance with the tool. Refer to section 3.5 of the PD for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable.
	Project owners and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	This is a small-scale project, hence TOOL19: “Demonstration of additionality of microscale project activities” is not applicable.

Tool27	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Project applies Tool19: “Demonstration of additionality of small-scale project activities”, v13.1. Hence this tool is applicable.
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	As per the methodology the additionality including investment analysis has been demonstrated as per the Tool 21: “Demonstration of additionality of small-scale project activities” version 13.1 and Tool 27: Investment Analysis, version 13.0 Hence Justified.
Tool20	This methodological tool is applicable to proposed small-scale project activities and small-scale CPAs in order to check whether they are debundled components of large-scale project activities.	This is a single 15 MW solar power project which comes under the small-scale project limit of 15 MW. Hence this tool is applicable.

3.3 Project Boundary

The project boundary includes the solar project, sub-stations, Indian grid and all power plants connected to the grid. The project activity will evacuate power through the Indian grid. Therefore, the entire Indian grid and all connected power plants have been considered in the project boundary for the project activity.

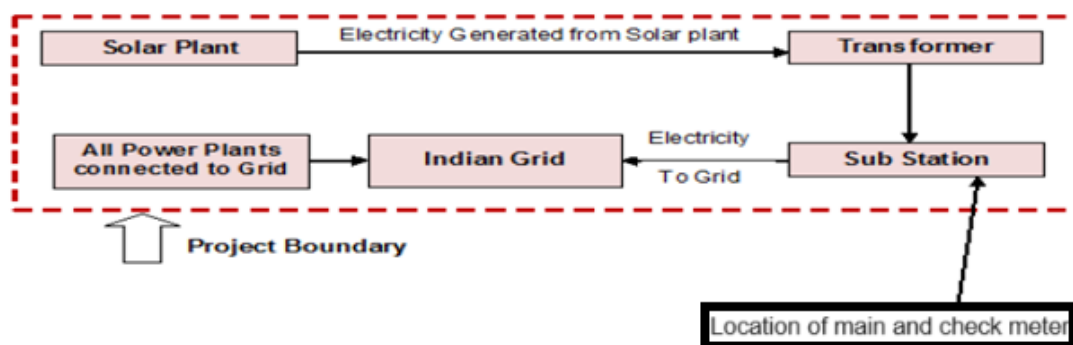


Figure: Project Boundary

The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main Emission Source
		CH ₄	No	Minor Emission source
		N ₂ O	No	Minor Emission source
		Other	-	-
Project	Greenfield Solar PV Power Project activity	CO ₂	No	Project activity does not emit CO ₂
		CH ₄	No	Project activity does not emit CH ₄
		N ₂ O	No	Project activity does not emit N ₂ O
		Other	-	-

3.4 Baseline Scenario

An approved small-scale baseline CDM methodology AMS-I.D. “Grid connected renewable electricity generation”, Version 18.0 has been followed along with the “Tool 07: tool to calculate the emission factor for an electricity system, version 7.0” are used to establish the baseline scenario.

According to the methodology baseline scenario is “Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07 “Tool to calculate the emission factor for an electricity system” Version 7.0 means that a power plant with emission factor equivalent to grid mix would have supplied electricity in the absence of new project plant or added capacity. A grid emission factor is a reasonable benchmark that provides the proxy performance of the baseline power plant.

The project involved setting up of a solar power generation plant to harness the power of solar energy to produce electricity and supply to the grid. In the absence of the project, the equivalent amount of power would have been supplied to the electricity grid by the operation of grid connected power plants and by addition of new generation sources, as reflected in the combined margin (CM) calculations.

Hence, the baseline for the project activity is the equivalent amount of electricity generated from the Indian grid connected power plants.

During the implementation of the project and establishing the baseline scenario the relevant national and/or sectoral policies, regulations and circumstances are taken into account.

- Implementation of solar energy-based power projects for electricity generation is not mandatory under any law in India, the project activity is a voluntary action.
- Despite the gradual increase in renewable energy sources (including solar energy) in power sector, still about two-third of installed power generation capacity is based on fossil fuel-based energy sources, hence the national grid is fed by electricity generated predominantly by the fossil-fuel based power plants.

There is no legal and regulatory requirement that mandates the production of energy by the chosen technology. Investment in solar energy projects in the State of Himachal Pradesh and the Indian electricity grid are not mandatory. There are no national or local laws or regulations that require this investment to be undertaken, i.e., setting up of solar power projects. The setting up of solar energy projects is a voluntary activity.

Project Owner has also considered the impact of E+ and E- policies while determining the baseline scenario by taking the guidelines as per paragraph 63-65 of CDM Project Standard consideration of relevant national and/or sectoral policies, regulations and circumstances shall be taken into account in the establishment of the baseline scenario⁸.

As per the referred guideline

- There is no such E+ policies have been implemented in the host country since adaptation of the Kyoto Protocol (11/12/1997) which give comparative advantage to increase the share of electricity in the national grid by connecting the more emission intensive fossil fuel-based power projects.
- In India there are certain policies have been implemented by Ministry of new and Renewable energy such as National Solar mission, Introducing Feed-in Tariffs Schemes, GBI (Generation Based Incentive) subsidy, Facilitating the developers through Viability Gap funding scheme and certain tax benefits etc. Despite having these policies to encourage the investments into the renewable energy sector, to increase the share of renewable energy installed capacity in the country, the share of fossil fuel-based electricity generation in the national grid accounts for more than 50% compared to the renewable energy technologies. Hence, the baseline scenario still remains valid as on date.
- In addition, there is no E- policy no particular technology is mandatory to use for the power generation purpose. It is neither mandatory to use the more intensive emission

⁸ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210921115752577/reg_stan04_v03.0.pdf

technologies nor less emission intensive technologies to generate the electricity power while supplying to the national grid. As explained above the project activity is a voluntary action.

Hence there is no impact of the E+ and E- policies while demonstrating the baseline scenario of this project activity.

In the absence of the project activity, the equivalent amount of electricity would have been drawn from the Indian grid. The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two Emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM). Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. The CEA database version 19.0 is the latest available, hence same is considered for emission factor calculations.

The combined margin emission factor of the Indian grid used for the project activity is as follows:

Parameter	Value	Nomenclature	Source
$EF_{grid,CM,y}$	0.9352 tCO ₂ /MWh	Combined margin CO ₂ emission factor for the project electricity system in year y	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Baseline CO ₂ Emission Database, Version 19.0 ⁹ published by Central Electricity Authority (CEA), Government of India
$EF_{grid,OM,y}$	0.9580 tCO ₂ /MWh	Operating margin CO ₂ emission factor for the project electricity system in year y	Calculated as the last 3 years (2020-21 ,2021-22& 2022-23,) generation-weighted average, sourced from Baseline CO ₂ Emission Database, Version 19.0, published by Central Electricity Authority (CEA), Government of India
$EF_{grid,BM,y}$	0.8670 tCO ₂ /MWh	Build margin CO ₂ emission factor for the project electricity system in year y	Baseline CO ₂ Emission Database, Version 19.0, published by Central Electricity Authority (CEA), Government of India

The baseline case is in compliance with all applicable legal and regulatory requirements references.

3.5 Additionality

The additionality of the project activity is demonstrated by following the guidance provided in the VCS project standard Version 4.6.

As per section 3.14 in VCS Standard v4.6, additionality can be demonstrated using the following components

⁹ https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☒ No

As per 3.14.1 demonstrating regulatory surplus, the Project conforms to all the applicable laws and regulations in India:

- Power generation using renewable energy is not a legal requirement or a mandatory option.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation
- Both the baseline and project activity are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity.
- The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations.

As per the section 3.14 of the VCS standard version 4.6, the project is not implemented by the force of law. This is a voluntary activity undertaken by the project owner in compliance with all the legal requirement in the host country. Hence project demonstrates regulatory surplus.

3.5.2 Additionality Methods

The project activity does not fulfil the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies”, version 4.0 and hence additionality of the project activity is demonstrated through a project specific additionality test.

For the demonstration and assessment of additionality, Tool 21 “Demonstration of additionality of small-scale project activities”, Version 13.1 has been applied. As per the para 10 of the tool, Project owner shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

Investment barrier: A financially more viable alternative to the project activity would have led to higher emissions.

Technological barrier: A less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;

Barrier due to prevailing practice: Prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;

Other barriers: Without the project activity, for another specific reason identified by the project owner, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

The project owner has selected Investment barrier to demonstrate in a conservative and transparent manner that the VCS project activity is financially unattractive.

To conduct the investment analysis, Tool 27: Investment analysis, version 13.0, EB 120 annex 3 have been referred.

Considering the fact that the alternative to the project is the supply of electricity from the grid & the choice of the developer is to invest or not to invest, benchmark analysis has been considered appropriate for demonstration of additionality, which is in conformity with “Tool 27: Investment Analysis”, version 13.0, EB 120 annex 3.

Apply benchmark analysis

According to the “Tool 01: Tool for demonstration and assessment of Additionality”, version 7.0.0, the financial indicator can be based either on (1) Project IRR or (2) Equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

As per Investment Analysis tool, required/expected returns on equity are appropriate benchmarks for a post- tax equity IRR. The post-tax equity IRR is considered as the financial indicator and the benchmarks used is cost of equity. Hence the benchmarks used are applicable to the project activity and the type of IRR calculation presented.

Selection of Appropriate Benchmark

This project needs to be demonstrating the additionality test based on the Project specific additionality test.

The benchmark has been considered in accordance with Guidance of Tool 27: Investment Analysis, version available at the time of investment decision making as well as the latest available value. As a conservative approach, the minimum value of benchmark has been considered as calculated using these 2 approaches.

As per para 19 of investment analysis, the cost of equity is determined by selecting the values provided in the Appendix, i.e. default values for cost of equity (expected return on equity) is presented below:

Default Value Benchmark:

As per Appendix in Tool 27: “Investment Analysis”, Version 12.0 the default value of expected return of equity in real terms for Energy Industries (Group 1) in India = 9.77%¹⁰ at the time of Investment decision.

Benchmark estimation:

The default value as per latest version of Tool 27: Investment analysis, version 13.0 is 9.13%¹¹ for group 1 project in India is used which is appropriate for benchmark calculation and the project owner has considered the same tool for default value of return on equity for the respective project.

Table under Appendix in EB120, Annex 3 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = **9.13%**

The Required return on equity (benchmark) was computed in the following manner:

There is no change in the cost of equity considered both at the time of investment decision and as per latest available version, therefore cost of equity considered for calculation of Benchmark = 9.13%

$$\text{Nominal Benchmark}^{12} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 9.13% (as per Appendix of EB 120 annex 3)

Thus, minimum cost of equity considered for calculation of Benchmark = 9.13%

Inflation Rate:

As per paragraph 16 of Appendix A of the above-mentioned document, “In situations where an investment analysis is carried out in nominal terms, project owners can convert the real term values provided in the table to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used”.

In line with investment analysis tool, Project owner has considered the next 5 years forecasted inflation rate published by the International Monetary Fund World Economic Outlook¹³ at the time of project investment decision (i.e., SJVN Financial Statement consideration) i.e., 13/01/2022. The applicable inflation rate and corresponding benchmark values are provided below for both the projects.

Inflation Forecast	Benchmark
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¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf>

¹² As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran

¹³ <https://www.imf.org/en/Publications/WEO/weo-database/2022/October/weo-report?c=534,&s=PCPI,PCPIPCH,&sy=2022&ey=2027&ssm=0&scsm=1&ssc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1>

4.32%	13.84%
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The benchmark of **13.84 %** has been considered for the investment analysis.

The Chronology of events of the project activity are provided below:

S. No.	Activity	Chronology of the project
1	Rfs date	03/07/2020
2	SJVN Pre-Bid Note	13/01/2022
3	Last date for Bid Submission	14/01/2022
4	Letter of Intent (LoI) to SJVN	04/10/2022
5	Power Purchase Agreement (PPA)	23/12/2022
6	LOA to EPC & O&M Contract	24/03/2023
7	Signing of EPC Contract	25/04/2023
8	Commissioning Date (Expected)	14/09/2024

Input values considered for computing investment analysis are given in below Table:

Calculation and comparison of financial indicators:

The Post Tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e., 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of IRR calculation.

The investment decision date of the project is 13/01/2022, which is the date of when SJVN signed the financial parameters for e-bidding process with committees' approval. Input values used in the investment analysis are sourced from the Pre-Bid note which is available at the time of the investment decision. The estimated PLF has been sourced from the "Energy Yield Assessment Report" prepared by third party engineering company which is also available at the time of investment decision making which is also in line with the guidelines of EB48 Annex 11.

Input values considered for the IRR calculation are provided below:

Particulars	Value	Unit	Source/Remarks
Capacity of the project	15	MW	Pre-Bid Note
Annual Net generation (First year)	31.150	GWh	PVsyst Report
Plant Load Factor	23.71%	%	Calculated
Annual Degradation	0.5%	%	Pre-Bid Note
Project cost	834.80	INR Million	Pre-Bid Note
Debt	80%	%	Pre-Bid Note
Equity	20%	%	
Debt	667.84	INR Million	Calculated
Equity	166.96	INR Million	Calculated
Interest rate	7.10%	%	Pre-Bid Note
Debt Repayment tenure	12	years	Pre-Bid Note
Moratorium	0	year	

Operation and Maintenance	14.2	INR Million	Pre-Bid Note
Escalation in O & M	3.84%	%	Pre-Bid Note
GST for O & M	18.00%	%	As per prevailing tax
Insurance & overhead	1.25	INR Million / Yr	Internal Assumption
Tariff	3.99	Rs/kWh	Pre-Bid Note
Gross Depreciable Value	834.80	INR Million	Calculated
Salvage Value	10%	%	
Salvage Value (@ 10%)	83.48	INR Million	Calculated
Net Depreciable Value	751.32	INR Million	Calculated
Residual Value	83.48	INR Million	Calculated
Depreciation Rate (Book)	4.00%	%	Calculated
IT Depreciation Rate	7.69%	%	As Per Income Tax, Depreciation rates for power generating units http://www.incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.html
Income tax rate	34.94%	%	Calculated
MAT rate	17.47%	%	Calculated
Project lifetime	25	years	Manufacturer specifications

Tax Rates

Financial Year	FY 2022-23		Source/Remarks
Income tax rate (%)	30.00%	Indian IT Act for FY 2022-23	https://www.incometax.gov.in/iec/foportal/help/company/return-applicable
MAT (%)	15.00%	Indian IT Act for FY 2022-23	
Surcharge (%)	12.00%	Indian IT Act for FY 2022-23	
Education cess (%)	4.00%	Indian IT Act for FY 2022-23	

Final Tax Rates

Income tax rate (%)	34.94%	%	Calculated
MAT (%)	17.47%	%	Calculated

Post Tax Equity IRR for the project activities against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the Post tax equity IRR is less below the benchmark value.

Post tax Equity IRR	10.36%
Benchmark Value	13.84%

Sensitivity Analysis

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 3 of EB 120, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. Project owner has identified the total revenue from the project activity is dependent on the Tariff, Net Annual Generation, Project Cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10%

variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to breach benchmark	Value required to breach benchmark
Tariff (INR/kWh)	7.34%	10.36%	13.51%	11.08%	4.432
Net Annual Generation (GWh)	7.34%	10.36%	13.51%	11.08%	34.60
Project Cost (INR (Mn))	13.23%	10.36%	8.13%	-11.88%	735.63
O&M Cost (INR (Mn))	10.92%	10.36%	9.76%	-61.46%	5.47268

An analysis has been done to identify the percentage variation at which the financial indicators will breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

- Tariff:** The Tariff rate of electricity used for investment analysis i.e., 3.99 INR/kWh is sourced from the Pre-Bid note available at the time of investment decision. Furthermore, the project will breach the benchmark value at a tariff variation of 11.08%. The Power Purchase agreement is signed for 25 years at fixed rate of 3.26 INR/kWh which is less than the estimated tariff, an increase in tariff is unlikely.
- Net Annual Generation:** The Net Annual Generation considered is based on the Third-Party pvsyst report i.e., 31.150 GWh and the Net Annual Generation breaches the benchmark value at variation of more than 11.08%. The chances of increase in generation value to breach the benchmark is very low as the considered generation is based on the energy yield assessment study.
- Project Cost:** The project cost considered for investment analysis i.e., 834.80 million INR. The cost is sourced from Pre-Bid note prepared by SJVN which is based on the negotiations with Supplier. A variation of -11.88% is required for IRR to breach benchmark.
- O&M Costs:** The Operation and Maintenance cost is considered from the SJVN Pre-Bid note available at the time of investment decision. Sensitivity analysis reveals that O&M will breach the benchmark at a variation of -61.46%. Since the O&M cost is subject to escalation and subject to inflationary pressure, any reduction in the O&M costs is highly unlikely.

Conclusion:

As described above, the project fulfills all necessary requirements of additionality specified in the Tool 21: “Demonstration of additionality of small-scale project activities”, version 13.1. Hence, the project is additional.

3.6 Methodology Deviations

This section is not applicable, as there is no deviation of methodology applied by the Project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

As per the methodology AMS-I.D: Grid connected renewable electricity generation, Version 18.0 emission reductions are calculated as follows

Emission Reductions

$$ER_y = BE_y - PE_y - LE_y$$

Where,

- ER_y = Emission reductions in year y (tCO₂)
- BE_y = Baseline emissions in year y (tCO₂)
- PE_y = Project emissions in year y (tCO₂)
- LE_y = Leakage emissions in year y (tCO₂)

4.1 Baseline Emissions

As per the applied methodology AMS-I.D.: Grid connected renewable electricity generation, Version 18.0, that Baseline emissions include only CO₂ emissions from electricity generation in grid connected power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where,

- BE_y = Baseline emissions in year y (tCO₂)
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
- $EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

As per paragraph 26 of AMS-I. D: Grid connected renewable electricity generation, version 18.0, when the project activity is installation of Greenfield power plant, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where,

- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{\text{facility}, y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The $EG_{\text{facility}, y}$ is estimated from the PLF provided as per the third party engineering company report.

The annual average electricity generated from the project with annual degradation is as below:

Project	$EG_{\text{facility}, y}$ (MWh)
Project (15 MW)	30,583

Calculating the Emission Reductions

As per the methodology combined margin grid emission factor has been calculated as per the “Tool 07: Tool to calculate the grid emission factor for an Electricity System”, version 7.0.

CO₂ Baseline Database for the Indian Power Sector, Version 19, November 2023¹⁴ published by Central Electricity Authority (CEA), Government of India has been used for the calculation of emission reduction.

As per the "Tool 07: Tool to calculate the emission factor for an electricity system" Version 7.0, the following steps have been followed.

- (a) **Step 1:** Identify the relevant electricity systems;
- (b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- (c) **Step 3:** Select a method to determine the operating margin (OM);
- (d) **Step 4:** Calculate the operating margin emission factor according to the selected method;
- (e) **Step 5:** Calculate the build margin (BM) emission factor;
- (f) **Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool “For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems”. It also states that “If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used”. Keeping this into consideration, the Central Electricity Authority (CEA), Government of India has divided the Indian Power Sector into five regional grids viz. Northern, Eastern, Western, North-eastern and Southern.

However, since August 2006, however, all regional grids except the Southern Grid had been integrated and were operating in synchronous mode, i.e., at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as NEWNE grid from FY 2007-08 onwards for the purpose of this CO₂ Baseline Database. As of 31 December 2013, the Southern grid has also been synchronized with the NEWNE grid; hence

¹⁴ https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations will serve as the baseline for this project.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project owners may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The Project owner has chosen only grid power plants in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

	Share of Must-Run (Hydro/Nuclear) (% of Net Generation)					
	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
India	14.3%	14.5%	17.0%	16.5%	15.8%	15.33%

Data Source: Central Electricity Authority (CEA) database Version 19, November '2023¹⁵

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

¹⁵ https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

- (a) **Ex-ante option:** if the ex-ante option is chosen, the emission factor is determined once at the registration stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

- (b) **Ex-post option:** if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

The project owner has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to VERRA.

OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWh) (incl. Imports)			
	2020-21	2021-22	2022-23
INDIAN Grid	958,218	1,035,672	1,117,846

Simple Operating Margin (tCO ₂ /MWh) (incl. Imports)			
	2020-21	2021-22	2022-23
INDIAN Grid	0.9402	0.9605	0.971

Weighted Generation Operating Margin	
INDIAN Grid	0.9580

STEP 5: Calculate the build margin emission factor ($EF_{BM,y}$)

Option 1 as described above is chosen to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.

Build Margin (tCO ₂ /MWh) (not adjusted for imports)	
	2021-22
INDIAN Grid	0.8670

STEP 6: Calculate the combined margin (CM) emissions factor

Combined Margin – The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, the Tool 07: Tool to calculate the emission factor for an electricity system, Version 7.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

The baseline emission factor is calculated using the combined margin approach as described in the following steps:

Calculation of Baseline Emission Factor $EF_{grid,CM,y}$

The baseline emission factor $EF_{grid,CM,y}$ is calculated as the weighted average of the Operating Margin emission factor ($EF_{grid,OM,y}$) and the Build Margin emission factor ($EF_{grid,BM,y}$):

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$$

Where,

WOM 75% weight for solar energy projects

WBM 25% weight for solar energy projects

$EF_{grid,OM,y}$ calculated as described in Steps 3 & 4 above (tCO₂/MWh)

$EF_{grid,BM,y}$ calculated as described in Steps 5 above (tCO₂/MWh)

$$\text{Baseline Emission factor (INDIAN Grid)} = 0.75 * 0.9580 + 0.25 * 0.8670$$

$$= 0.9352 \text{ tCO}_2/\text{MWh}$$

The baseline emission factor is ex-ante parameter and will remain constant throughout the crediting period.

$EF_{grid,,} =$ Combined Margin Grid Emission Factor = 0.9352 tCO ₂ /MWh

4.2 Project Emissions

As per the approved consolidated Methodology AMS-I.D (Version 18.0) para 39: For most renewable energy power generation project activities, $PE_y = 0$.

As the project activity is the installation of a new grid-connected solar power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants.

Therefore $PE_y = 0$.

4.3 Leakage Emissions

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g: extraction, processing, transport etc.) are neglected.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The ex-ante emission reductions (ER_y) for the project activity are calculated as follows

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y = Emission Reduction in year y (tCO_2)

BE_y = Baseline emission in year y (tCO_2)

PE_y = Project emissions in year y (tCO_2)

LE_y = Leakage Emissions in year y (tCO_2)

Baseline Emissions (BE_y):

The baseline emissions are the product of electrical energy baseline $EG_{PJ,y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor.

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

As per para 26 of AMS I.D, version 18.0, when the project activity is installation of Greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ, facility,y}$$

Where,

$EG_{PJ, facility,y}$ = Total quantity of net electricity delivered to the INDIAN grid in year y (MWh)

$EF_{grid,y}$ = Baseline grid emission factor (tCO_2/MWh)
= 0.9352 tCO_2/MWh

The $EG_{PJ, facility,y}$ is sourced from the energy yield assessment report provided by the third-party engineering company. The annual estimated electricity generation is provided below for the total crediting period:

Project	$EG_{PJ, facility,y}$ (MWh)
Project (15 MW)	30,583

As per section 4.1 above, the combined margin grid emission factor ($EF_{grid,CM,y}$) is 0.9352 tCO_2/MWh

Hence the annual baseline emission is calculated as below:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} = 30,583 \text{ MWh} \times 0.9352 \text{ } tCO_2/MWh = \mathbf{28,601 \text{ } } tCO_2$$

Project Emissions (PE_y):

As explained in the above section B.6.1, Project emissions from the project activity is considered Zero.

$$PE_y = 0$$

Leakage Emissions (LE_y):

As explained in the above section B.6.1, Leakage emissions from the project activity is considered zero.

$$LE_y = 0$$

Hence, Net Emission reductions will be calculated as per the below equation

$$ER_y = BE_y - PE_y - LE_y$$

Considering the commissioning date of the project and annual degradation, the emission reduction estimation for the entire crediting period is provided in the below section.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
14-Sep-2024 to 31-Dec-2024	8,700	0	0	8,700	0	8,700
01-Jan-2025 to 31-Dec-2025	28,989	0	0	28,989	0	28,989
01-Jan-2026 to 31-Dec-2026	28,844	0	0	28,844	0	28,844
01-Jan-2027 to 31-Dec-2027	28,700	0	0	28,700	0	28,700
01-Jan-2028 to 31-Dec-2028	28,556	0	0	28,556	0	28,556
01-Jan-2029 to 31-Dec-2029	28,414	0	0	28,414	0	28,414
01-Jan-2030 to 31-Dec-2030	28,272	0	0	28,272	0	28,272
01-Jan-2031 to 13-Sep-2031	19,729	0	0	19,729	0	19,729
Total	200,204	0	0	200,204	0	200,204

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor in year y of Indian Grid.
Source of data	Calculated from CEA database, Version 19, November 2023 ¹⁶

¹⁶ https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

Value applied	0.9580										
Justification of choice of data or description of measurement methods and procedures applied	<table border="1"> <tr> <td colspan="2">Not Applicable</td></tr> <tr> <td>Type of meter</td><td>Not Applicable</td></tr> <tr> <td>Location of meter</td><td>Not Applicable</td></tr> <tr> <td>Accuracy of meter</td><td>Not Applicable</td></tr> <tr> <td>Serial number of meter</td><td>Not Applicable</td></tr> </table> Calculated in line with “Tool to calculate the emission factor for an electricity system” version 7.0 using data from Central Electricity Authority of India’s (CEA) “Baseline Carbon Dioxide Emission Database Version 19.0”. The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the CEA database. Weighted average = $\sum_{i=1 \text{ to } n} (\text{Net generation in operating margin in year } i * \text{Simple operating margin in year } i) / \sum_{i=1 \text{ to } n} (\text{Net generation in operating margin of year } i)$	Not Applicable		Type of meter	Not Applicable	Location of meter	Not Applicable	Accuracy of meter	Not Applicable	Serial number of meter	Not Applicable
Not Applicable											
Type of meter	Not Applicable										
Location of meter	Not Applicable										
Accuracy of meter	Not Applicable										
Serial number of meter	Not Applicable										
Purpose of data	Baseline Emission calculation										
Comments	The operating margin emission factor is a 3-year generation-weighted average (2020-23). The operating Margin is calculated ex ante and fixed during the crediting period										

Data / Parameter	$EF_{\text{grid,BM},y}$										
Data unit	tCO ₂ /MWh										
Description	Build Margin CO ₂ emission factor in year y of Indian Grid.										
Source of data	Calculated from CEA database, Version 19, November 2023										
Value applied	0.8670										
Justification of choice of data or description of measurement methods and procedures applied	<table border="1"> <tr> <td colspan="2">Not Applicable</td></tr> <tr> <td>Type of meter</td><td>Not Applicable</td></tr> <tr> <td>Location of meter</td><td>Not Applicable</td></tr> <tr> <td>Accuracy of meter</td><td>Not Applicable</td></tr> <tr> <td>Serial number of meter</td><td>Not Applicable</td></tr> </table> Calculated in line with “Tool to calculate the emission factor for an electricity system” version 7.0 using data from Central Electricity Authority of India’s (CEA) “Baseline Carbon Dioxide Emission Database Version 19.0”. The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the CEA database.	Not Applicable		Type of meter	Not Applicable	Location of meter	Not Applicable	Accuracy of meter	Not Applicable	Serial number of meter	Not Applicable
Not Applicable											
Type of meter	Not Applicable										
Location of meter	Not Applicable										
Accuracy of meter	Not Applicable										
Serial number of meter	Not Applicable										

Purpose of Data	Baseline Emission calculation
Comments	The build margin emission factor is a 3-year generation-weighted average (2022-23). The operating Margin is calculated ex ante and fixed during the crediting period

Data / Parameter	EF _{grid,CM,y}										
Data unit	tCO ₂ /MWh										
Description	Combined Margin CO ₂ emission factor in year y of Indian Grid.										
Source of data	Calculated from CEA database, Version 19, November 2023										
Value applied	0.9352										
Justification of choice of data or description of measurement methods and procedures applied	<table border="1"> <tr> <td colspan="2">Not Applicable</td></tr> <tr> <td>Type of meter</td><td>Not Applicable</td></tr> <tr> <td>Location of meter</td><td>Not Applicable</td></tr> <tr> <td>Accuracy of meter</td><td>Not Applicable</td></tr> <tr> <td>Serial number of meter</td><td>Not Applicable</td></tr> </table> The date has been considered in accordance to the Tool to calculate emission factor of an electricity system. The tool guides to take 75% weightage of EF_{grid}, OM_{simple}, & 25% weightage of EF_{grid,BM,y}.	Not Applicable		Type of meter	Not Applicable	Location of meter	Not Applicable	Accuracy of meter	Not Applicable	Serial number of meter	Not Applicable
Not Applicable											
Type of meter	Not Applicable										
Location of meter	Not Applicable										
Accuracy of meter	Not Applicable										
Serial number of meter	Not Applicable										
Purpose of Data	Baseline Emission calculation										
Comments	The combined margin would be calculated ex-ante and fixed for the entire crediting period and the combined margin thus calculated is 0.9352.										

5.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project (Solar) plant/unit to the grid in year y
Source of data	JMR from End user (consumer)
Description of measurement methods and procedures to be applied	Measured & calculated

Frequency of monitoring/recording	Monthly recording																													
Value applied	30,583																													
Monitoring equipment	<table><tr><td>Project 1</td><td>Main Meter</td><td>Check meter</td></tr><tr><td>Type of meter</td><td colspan="2">Trivector Bi-directional meters (Manufacturer: Secure Meters Limited)</td></tr><tr><td>Location of meter</td><td colspan="2">BBMB Substation</td></tr><tr><td>Accuracy of meter</td><td colspan="2">0.2s</td></tr><tr><td>Serial number of meters</td><td colspan="2">To Be Provided</td></tr><tr><td>Calibration frequency</td><td colspan="2">To Be Provided</td></tr><tr><td>Date of Calibration/ validity</td><td colspan="2">To Be Provided</td></tr><tr><td>Reference No. of Calibration Certificate</td><td colspan="2">To Be Provided</td></tr><tr><td>Calibration Status</td><td colspan="2">To Be Provided</td></tr></table>			Project 1	Main Meter	Check meter	Type of meter	Trivector Bi-directional meters (Manufacturer: Secure Meters Limited)		Location of meter	BBMB Substation		Accuracy of meter	0.2s		Serial number of meters	To Be Provided		Calibration frequency	To Be Provided		Date of Calibration/ validity	To Be Provided		Reference No. of Calibration Certificate	To Be Provided		Calibration Status	To Be Provided	
	Project 1	Main Meter	Check meter																											
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	Date of Calibration/ validity	To Be Provided																												
	Reference No. of Calibration Certificate	To Be Provided																												
	Calibration Status	To Be Provided																												
Energy meters of accuracy class 0.2s.																														
Main and Check meters are installed at the BBMB substation by the electricity utility to measure the net exported electricity from the plant.																														
QA/QC procedures to be applied	The meter(s) shall be calibrated and maintained by the DISCOMs as per their own schedule, and this frequency of meter calibration is not within the control of the Project Owner. Calibration of electricity meters is carried out in-line with the National standard¹⁷ which recommends at least once in 5-year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter.																													
Purpose of data	Baseline Emission Calculations																													
Calculation method	The value of net electricity generation supplied to the grid as per Monthly Joint Meter Reading Report forms the basis for calculation of the emission reductions, which can be cross-checked from the invoice raised to Consumer. The Net electricity is calculated based on Export - import.																													

¹⁷ (Page number 18 of https://cea.nic.in/old/reports/regulation/CEA_metering_regulation_amendment_2019.pdf)

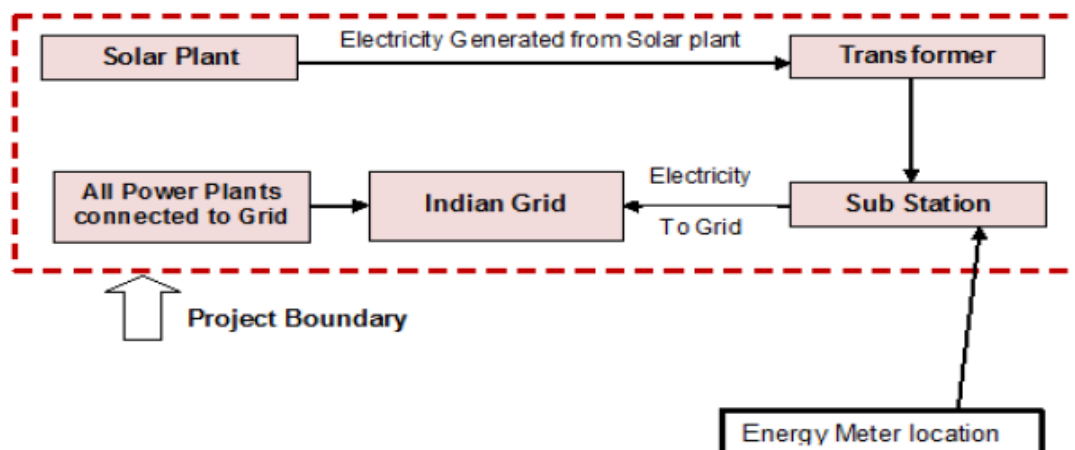
	Monthly meter readings are taken from the main and check meter installed at metering point and certified by the representatives of DISCOM Officials and the representatives of the project owner participant. The export and import values of the monthly Joint Meter Reports is cross checked with the export and import values mentioned in the invoice.
Comments	Monitored data will be stored for at least two years after the end of crediting period or the last issuance of for this project activity, whichever occurs later

5.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures of VERRA and is proposed for grid-connected solar power project being implemented in Himachal Pradesh, India. The monitoring plan, which will be implemented by the project owner describes about the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

The project owner has entered into agreement with the O&M Contractor for the operation and maintenance of Solar Plant. O&M contractor will provide a monthly report, which includes generation data, major breakdown events and machine availability. Project manager is responsible for recording of monthly meter readings of export and import. Monthly power export and import data will be sent regularly to site in charge of each project separately.

The electricity generated from the project activity will be supplied to the national grid for the complete project lifetime under a long-term PPA with BBMB. The electricity generated from the project activity before entering the grid at the connection point will be measured by digital kilowatt hour (kWh) meters. The power meters at the connection point are bi-directional nature which records the net export and import. Both main and check meters of +/-0.2s accuracy class have been installed at the substation. These are the 3-phase bi-directional trivector meters.



The steps of monitoring the electricity supplied to the national grid and the electricity imported from grid by the project are as follows:

1. The electricity supplied by the project to the grid and electricity imported from the grid are measured automatically by the bi-directional meter systems (main and backup power meters). The data is measured continuously
2. Persons in charge of data record and meter supervisor from the Project owner together with staff from DISCOM shall read and collect data from power meters on the first day of every month, the result will be signed by both parties and kept respectively.
3. The data from the backup power meters will be cross checked with the data from main power meter. The data from back-up system will be used in case of failing of the main meter;
4. The Project owner provides the record of main, backup power meters, copy of invoices and other related documents to the verifier.

Personal Training: The project employed qualified and experienced persons for plant operation. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Data recording & archiving: The project owner shall maintain data both in electronic form and hard copies.

Emergency preparedness:

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible.

- In normal condition, the data of main power meter will be used as the basis of payment by the DISCOM to the project owner and to calculate the emission reductions by the project activity.
- In case the main power meter is in failure and the backup power meter is still in good operation, the result of the backup meter will be used to calculate the emission reduction by the project activity.
- In case of both main and backup power meters are in failure, the Project Owner and the power company (DISCOM) will jointly calculate a conservative estimate of power supplied to the grid. The assumptions used to estimate net electricity supply to the grid will be signed by both a representative of the project owner as well as a representative of the power company (DISCOM).
- In case any power meters are in failure, the Project Owner will inform DISCOM immediately and contract with the authorized party (belong to DISCOM) to verify/ calibrate and/or replace the failed power meter.

Internal auditing

The main energy meter readings are verified against the reading of the backup meter to check any considerable variation. In case, considerable variation found, then either main meter reading or check meter reading whichever lower will be considered. The electricity readings recorded are

used for electricity invoicing to consumer. Hence, the data collected is of high accuracy and authorized by DISCOM.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

There is no sensitive information.

Section	Information	Justification
Not Applicable	Not Applicable	Not Applicable